Evaluating and Training Large Language Models: What Can (and Cannot) Item Parameters Tell Us?

Research Group Seminar · October 26

Central Question

Can static item parameters (a_j, b_j) calibrated across leaderboard LLMs guide adaptive evaluation and reinforcement learning—and when they fail, what can item-level rollout distributions tell us instead?

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1. The Evaluation & RL Gap

Outcome RL (`GRPO`) beats `SFT` on single-draw $\mathrm{Pass@1}$, yet collapses on $\mathrm{Pass@16}$ and adds **zero net gain** on Majority Vote ($\mathrm{Maj@16}$). Plugging leaderboard 2PL into adaptive testing (`CAT`) or RL critics hits a structural ceiling.

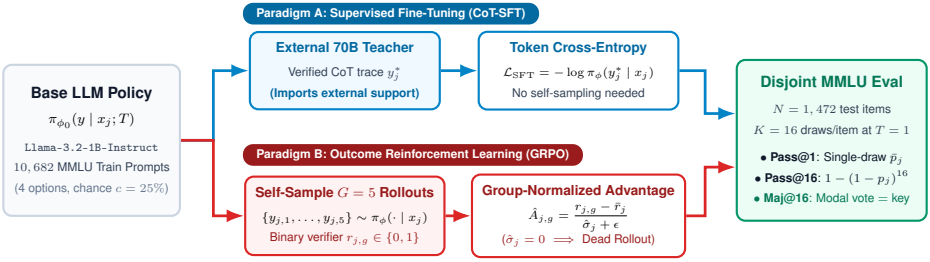
2. Rollout Diagnostics

Without retraining, multi-draw rollouts separate **policy sharpening** ($\tau \leftrightarrow 1/T$) from **distractor overthrow**, exposing why RL locks in wrong modal beliefs while wasting over half of active updates in the $\leq 25\%$ guessing region.

3. Dynamic Capability Tracking

Static standing $\neq$ gradient plasticity ($r = 0.072$). Guiding multi-step training requires a state-space formulation that separates epistemic filtering from answer-blind plasticity via an asymmetric teach–measure kernel.

Post-Training Paradigms and Multi-Draw Evaluation



CoT-SFT (Teacher Imitation)

Minimizes $-\log \pi_\phi(y_j^* \mid x_j)$ on verified 70B reasoning traces. Injects external support even when Base pass rate $p_j = 0$.

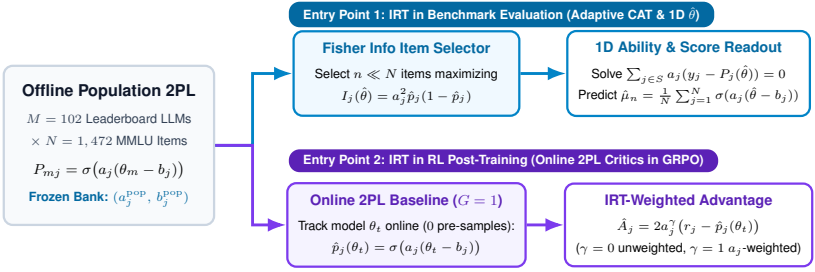
GRPO (Binary Outcome RL)

Rewards $\{0, 1\}$ correctness across $G = 5$ self-rollouts ($\hat{\sigma}_j = \sqrt{p_j(1 - p_j)}$). Blind to the 25% guessing floor and which wrong distractor is modal.

Multi-Draw Readouts ($K = 16$)

- $\mathbf{Pass@1}$: Mean single-draw accuracy.
- $\mathbf{Pass@16}$: Coverage across 16 draws.
- $\mathbf{Maj@16}$: Whether the modal option matches the answer key ($p_{\mathrm{key}} = p_{\mathrm{top}}$).

How Static Item Parameters Enter LLM Pipelines



Adaptive Evaluation (CAT)

Calibrate 1D binary 2PL (a_j, b_j) across $M = 102$ leaderboard LLMs (Maia Polo et al., 2024; Kipnis et al., 2025), then select items maximizing Fisher information $I_j(\hat{\theta}) = a_j^2 \hat{p}_j(1 - \hat{p}_j)$.

Online 2PL Critics in RL

Replace $G = 5$ group rollouts with a single rollout ($G = 1$) using static 2PL expectation $\hat{p}_j(\theta_t) = \sigma(a_j(\theta_t - b_j))$:

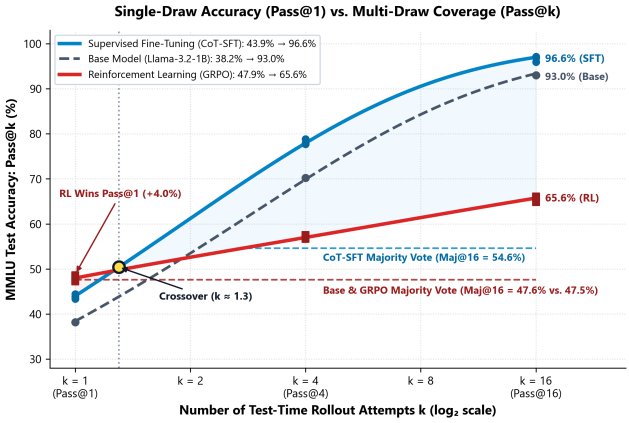
$$\hat{A}_j = 2\, a_j^{\gamma}\big(r_j - \hat{p}_j(\theta_t)\big), \qquad \gamma \in \{0, 1\}$$

Connecting Rollout Statistics to Psychometric Theory

Dimension	LLM Rollout Statistic	Psychometric Construct	Diagnostic Metric
1. Item Baseline	**Base initial pass rate ($p_{\text{base},j}$)**	**Model-specific item log-odds (μ_{0j})**	$\mu_{0j} = \text{logit}(p_{\text{base},j})$
2. Sharpening	**Inverse temperature ($1/T$) or RL**	**Within-model dispersion slope (τ)**	$\tau \leftrightarrow 1/T$
3. Consistency	**Easy-item misses vs. step curve**	**Drasgow Person-Fit & Guttman order**	$l_z = +7.55$ (RL) vs. -16.99 (
4. Guessing (c)	**Active GRPO variance at $\leq 25\%$**	**2PL Variance vs. 3PL Information**	$I_j^{\text{3PL}} = 0$ at $p_j \leq 25\%$
5. Distractor Mode	**Modal option ($p_{\text{top},j}$) vs. key ($p_{\text{key},j}$)**	**Nominal Response Model (Bock, 1972)**	$p_{\text{top},j} > p_{\text{key},j}$ (51.4% of ite
6. Prompt Bias	**Option-order gap (`A/B/C/D` swap)**	**Differential Item Functioning (DIF)**	$D = \lvert p_{\text{ord0}} - p_{\text{ord1}} \rvert$

Core Idea: Without retraining, item-level rollout statistics ($\mu_{0j}, \tau, l_z, I_j^{\text{3PL}}, p_{\text{top},j}$) diagnose *why* single-draw evaluation and static 2PL fail—and how `SFT` and `RL` play complementary roles.

The Pass@1 vs. Pass@k Evaluation Paradox



1. The Single-Draw Illusion ($k = 1$)

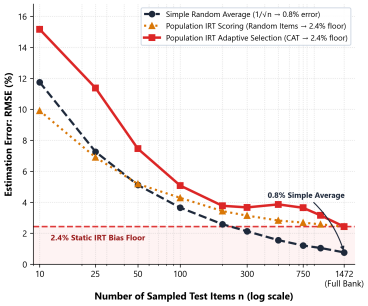
- Base: $\mathbf{38.2}\%$ | CoT-SFT: $\mathbf{43.9}\%$
- GRPO (RL): $\mathbf{47.9}\%$ ($+4.0\%$ ahead of SFT on single-draw accuracy).

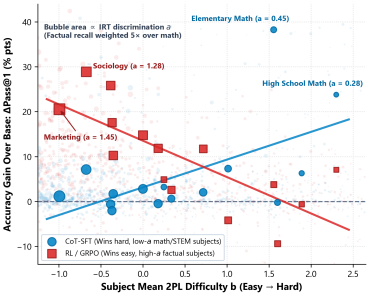
2. Reversal on Coverage & Majority Vote

- **Coverage collapse**: At $k = 16$, GRPO drops to $\mathbf{65.6}\%$ vs. Base $\mathbf{93.0}\%$ and CoT-SFT $\mathbf{96.6}\%$.
- **Zero modal gain under RL**: GRPO $\mathrm{Maj@16}$ ($\mathbf{47.5}\%$) matches Base ($\mathbf{47.6}\%$), whereas CoT-SFT reaches $\mathbf{54.6}\%$.

$\rightarrow$ RL sharpens existing Base modes; SFT flips modal answers to the correct key.

Why Adaptive Testing Biases Toward Memorization





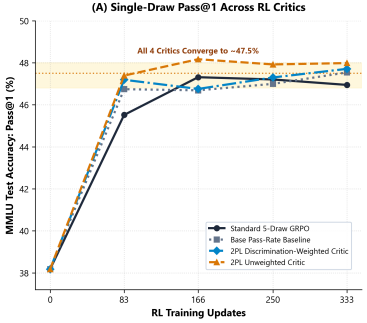
(A) Adaptive CAT Hits a 2.4% Bias Floor

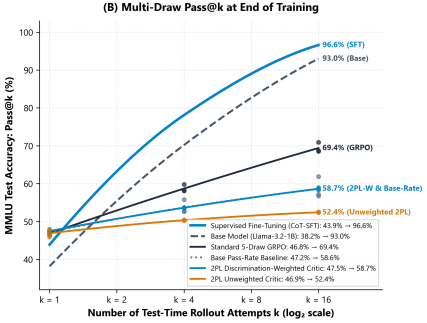
Simple random sampling reaches $\mathbf{0.8}\%$ RMSE at full bank, whereas 1D IRT scoring and CAT plateau at $\mathbf{2.4}\%$ error because post-trained LLMs violate 1D item invariance.

(B) $5\times$ Discrimination Skew Across Domains

Factual recall has $5\times$ higher a_j (*Marketing* $\mathbf{1.45}$) than reasoning (*HS Math* $\mathbf{0.28}$). Since $I_j \propto a_j^2$, CAT over-samples memory $25\times$, favoring RL over SFT.

Why Static 2PL Critics Saturate in RL





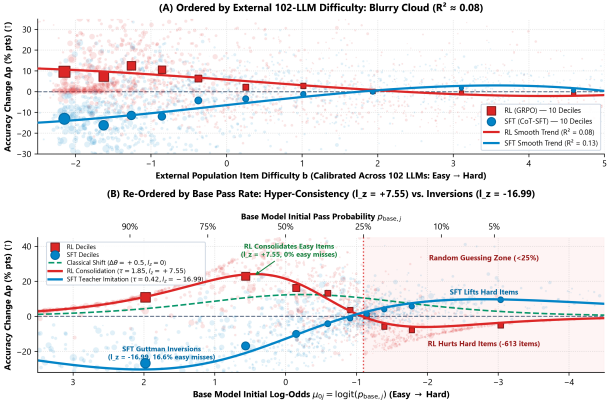
(A) All Four Critics Converge to $\sim 47.5\%$ Pass@1

Standard `GRPO` ($G = 5$) and single-rollout ($G = 1$) 2PL critics all plateau at $\mathbf{46.8\%}$–$\mathbf{47.5\%}$. Online 2PL cuts rollout cost $5\times$, but static a_j-weighting adds **zero accuracy gain**.

(B) High-k Collapse & Item-Subset Divergence

At $k = 16$, all RL critics collapse ($\mathbf{52.4\%}$–$\mathbf{69.4\%}$) below `Base` ($\mathbf{93.0\%}$). Despite tied aggregate Pass@1, **5 of 6 critic pairs** learn statistically distinct item subsets ($p < 0.05$).

From Leaderboard Standing to Within-Model Plasticity



1. Standing $\neq$ Plasticity (Panel A)

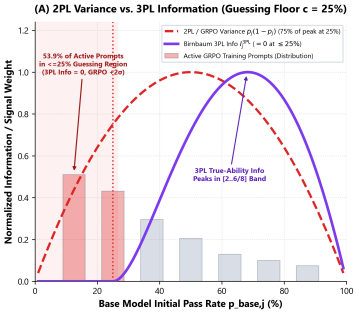
Ordering item gains Δp_j by 102-LLM difficulty b_j yields a flat cloud ($\mathbf{R^2 = 0.08}$). Within-model gradient updates are orthogonal to cross-model a_j ($\mathbf{r = 0.072}$).

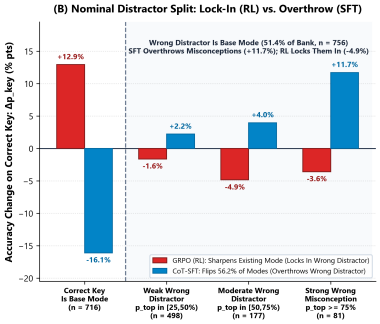
2. Base Frame (μ_{0j}) & Person-Fit l_z (Panel B)

Ordering by Base log-odds $\mu_{0j} = \text{logit}(p_{\text{base},j})$ ($\mathbf{R^2 = 0.35}$) reveals opposite consistency regimes:

- **RL Consolidation ($\tau = 1.85, l_z = +7.55$)**: Step-like Guttman order (0% easy misses), but degrades hard items ($< 25\%$).
- **SFT Imitation ($\tau = 0.42, l_z = -16.99$)**: Lifts hard items, but introduces 16.6% easy-item inversions.

3PL Guessing and Attractive Distractors





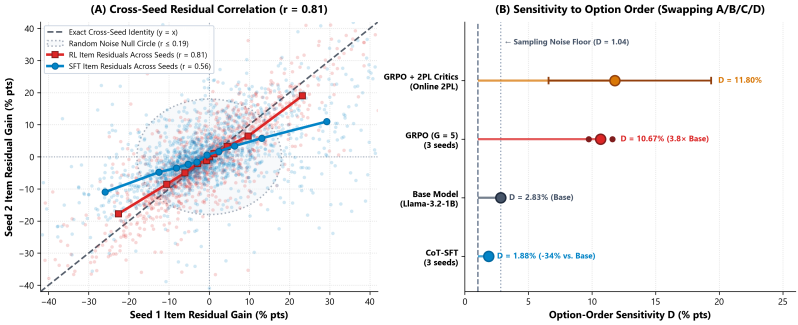
(A) 3PL Guessing Floor ($I_j^{\mathrm{3PL}} = 0$ at $\leq 25\%$)

At the $c = 25\%$ guessing floor, Birnbaum's 3PL information is $\mathbf{0}$, yet binary variance $p_j(1-p_j)$ remains $\mathbf{75}\%$ of peak. Over half ($\mathbf{53.9}\%$) of active GRPO prompts sit in $\leq 25\%$, reinforcing random guessing.

(B) Nominal Distractor Split ($p_{\mathrm{top},j} > p_{\mathrm{key},j}$)

On $\mathbf{51.4}\%$ of items, a wrong distractor is Base's mode ($\bar{p}_{\mathrm{top}} = \mathbf{48.0}\% > \bar{p}_{\mathrm{key}} = \mathbf{19.6}\%$). **RL locks in wrong modes** ($-\mathbf{4.9}\%$), whereas **CoT-SFT flips $\mathbf{56.2}\%$ of modes** ($+\mathbf{11.7}\%$ on strong misconceptions).

Cross-Seed Residual Structure and Option-Order Bias



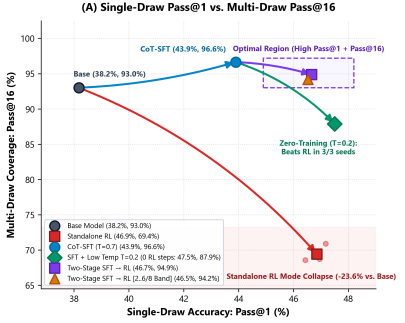
(A) Cross-Seed Residual Correlation ($\mathbf{r = 0.81}$)

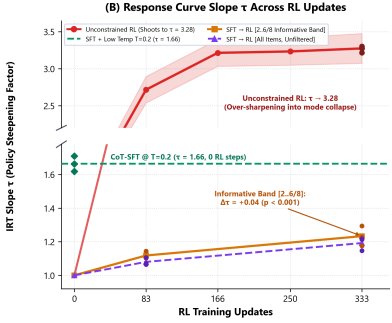
After removing the 1D curve, item residuals across random seeds remain strongly correlated ($\mathbf{r = 0.81}$ vs. 0.19 noise floor), showing stable multi-dimensional skill structure.

(B) RL Amplifies Option-Order Bias $4\times$

Swapping option letters (`A/B/C/D`) shows `CoT-SFT` reduces position bias ($\mathbf{1.88}\%$ vs. Base $\mathbf{2.83}\%$), whereas `GRPO` amplifies position bias $4\times$ to $\mathbf{10.67}\%$ when sharpening low-confidence items.

Actionable Fixes Without Full Retraining





1. Temperature Scaling

Since RL acts as $\tau \leftrightarrow 1/T$, decoding `CoT-SFT` at $\mathbf{T = 0.2}$ (**0 RL steps**) matches RL $\mathrm{Pass@1}$ ($\mathbf{47.5}\%$) while preserving $\mathbf{87.9}\%$ $\mathrm{Pass@16}$.

2. Two-Stage `SFT` $\rightarrow$ `RL`

`SFT` first overthrows wrong modal distractors ($p_{\mathrm{top}} > p_{\mathrm{key}}$); `RL` then repairs SFT's easy-item inversions ($l_z = -16.99 \rightarrow \mathbf{94.9}\%$ $\mathrm{Pass@16}$).

3. 3PL Band $[2..6/8]$

Filtering out the $1/8$ guessing bin ($I_j^{3\mathrm{PL}} = 0$) cuts dead RL rollouts $\mathbf{34}\% \rightarrow \mathbf{23}\%$ and steepens the consolidation slope ($+\mathbf{0.04}, \mathbf{p < 0.001}$).

Summary and Key Takeaways

1. Evaluation & Training Gaps

• **Single-Draw Illusion**: $\mathrm{Pass@1}$ confounds policy sharpening ($\tau \leftrightarrow 1/T$) with capability growth, masking RL's $\mathbf{0}\%$ modal gain ($\mathrm{Maj@16} = 47.5\%$) and $-\mathbf{23.6}\%$ coverage drop.

• **Binary RL Blindness**: Outcome RL (`GRPO`) wastes $\mathbf{53.9}\%$ of active updates in the $\leq 25\%$ guessing region ($I_j^{\mathrm{3PL}} = 0$) and locks in wrong modal distractors ($p_{\mathrm{top}} > p_{\mathrm{key}}$ on $\mathbf{51.4}\%$ of items).

2. Why Static 2PL Readouts Fail

• **Domain Skew in `CAT`**: Leaderboard a_j weights factual recall $5\times$ higher than math (1.45 vs. 0.28), creating a $\mathbf{2.4}\%$ adaptive testing bias floor.

• **Standing $\neq$ Plasticity**: Cross-model a_j is orthogonal to within-model gradient updates ($\mathbf{r = 0.072}$); all static 2PL RL critics saturate at $\sim \mathbf{47.5}\%$.

3. What Rollout Diagnostics Enable

• **Complementary Regimes**: Separates **RL Guttman Consolidation** ($\tau = 1.85, l_z = +\mathbf{7.55}$) from **SFT Distractor Overthrow** ($\mathbf{56.2}\%$ mode flips, $l_z = -\mathbf{16.99}$).

• **Immediate Gains**: Enables `CoT-SFT @ T=0.2` ($\mathbf{47.5}\%$ $\mathrm{Pass@1}$ in 0 RL steps), `Two-Stage SFT→RL` ($\mathbf{94.9}\%$ $\mathrm{Pass@16}$), and `3PL Band` $[2..6/8]$ ($\mathbf{34}\% \rightarrow \mathbf{23}\%$ dead rollouts).

Why Static Snapshots Cannot Guide Dynamic Training

1. Non-Ergodicity: Cross-Model Standing $\neq$ Within-Model Gradient Velocity

$$a_j^{\mathrm{dyn}}(t) \triangleq \frac{d\,\mathrm{logit}\,P_j(\boldsymbol{\phi}_t)}{dt} = \underbrace{\nabla_{\boldsymbol{\phi}}\mathrm{logit}\,P_j(\boldsymbol{\phi}_t)^\top}_{\text{Item Jacobian } \boldsymbol{J}_j} \underbrace{\dot{\boldsymbol{\phi}}_t}_{\text{Update Velocity } \boldsymbol{u}_t} \perp a_j^{\mathrm{pop}} \quad (r = 0.072)$$

Static rollout diagnostics ($\hat{p}_{j,0}, l_z, I_j^{\mathrm{3PL}}$) are single-checkpoint snapshots. Across training steps, logit evolution combines **global sharpening (τ_t)**, **global drift (δ_t)**, and **local support shift ($\Delta\kappa_{t,j}$)**.

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2. The Second-Order $\mathcal{O}(\Delta^2)$ Sink Principle

$$\hat{\theta}_W - \theta_W^* \;\approx\; \mathcal{J}_{W\cdot o}^{-1} \sum_{t=1}^{T} \mathbb{E}\big[p_t(1-p_t)\,\delta_t\,g_t\big]$$

Step-to-step skill transitions enter binary outcomes at **second order** $\mathcal{O}(\Delta^2)$, acting as a statistical sink for any unmodeled $\mathcal{O}(1)$ variation (temperature τ_t, baseline standing, guessing floor). Identifying transitions requires **closing $\mathcal{O}(1)$ nuisances first** and evaluating over **multi-step open-loop horizons ($H > 1$)**.

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3. Subspace Absorption & Read–Write Asymmetry

$$\mathcal{P}_\perp(\boldsymbol{W}) \;=\; (\boldsymbol{I} - \boldsymbol{A}\boldsymbol{A}^+)\boldsymbol{W}, \qquad \boldsymbol{a}_j^{\mathrm{read}} \;\neq\; \boldsymbol{\psi}_j^{\mathrm{write}}$$

Unconstrained global trajectories absorb transitions inside $\mathrm{col}(\boldsymbol{A})$. Moreover, forcing a single item vector to serve as both **static read key ($\boldsymbol{a}_j^{\mathrm{read}}$)** and **dynamic write vector ($\boldsymbol{\psi}_j^{\mathrm{write}}$)** collapses multi-skill clusters into item memorization.

A State-Space Model of Training Dynamics

1. Hierarchical Observation Model & Doob Martingale Decomposition

$$P_{j,t} = \sigma\Big(\tau_t\big(\boldsymbol{m}_j^\top \boldsymbol{v}_t + \boldsymbol{a}_j^\top \boldsymbol{z}_t + \boldsymbol{u}_j^\top \boldsymbol{c}_t - b_j\big)\Big), \qquad \Delta\boldsymbol{\kappa}_t = \underbrace{\boldsymbol{K}_t(y_t - \bar{p}_t)}_{\text{Epistemic Filter } \Delta\boldsymbol{z}_t \ (\mathbb{E}[\Delta\boldsymbol{z}_t]=\mathbf{0})} + \underbrace{\boldsymbol{g}_t(\bar{p}_t, \boldsymbol{\psi}_{e_t})}_{\text{Answer-Blind Drift } \Delta\boldsymbol{c}_t}$$

Frozen read keys $\boldsymbol{a}_j$ are identified from static residual covariance ($r = 0.81$). Two structural zeros separate **epistemic filtering** ($\Delta\boldsymbol{z}_t$, zero drift on frozen checkpoints) from **answer-blind plasticity** ($\Delta\boldsymbol{c}_t$, which never reads y_t).

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2. 3PL Interior Plasticity Law & Preconditioned State Transition

$$\mathrm{Cov}_{\pi_\phi}\big(r,\, \nabla_\phi \log \pi_\phi\big) = p_j(1-p_j)\boldsymbol{\psi}_j \implies \boldsymbol{c}_{t+1} = (\boldsymbol{I} - \boldsymbol{\Lambda})\,\boldsymbol{c}_t + \underbrace{c\,(\bar{p}_t - c_0)_+^{k_1}(1-\bar{p}_t)^{k_2}}_{\text{3PL Interior Gate } u(\bar{p}_t)}\,(1+\boldsymbol{s}_t)^{-\beta} \odot \boldsymbol{\psi}_{e_t}$$

Unifies SGD ($k = 1$), std-normalized RL ($k = \frac{1}{2}$), and the 3PL guessing floor $c_0 = 0.25$ ($[2..6/8]$ rollout band), while $(1+\boldsymbol{s}_t)^{-\beta}$ ($\boldsymbol{s}_{t+1} = \boldsymbol{s}_t + \boldsymbol{g}_t^2$) captures second-moment saturation and $\boldsymbol{\Lambda} \in (0,1)^M$ controls consolidation timescales.

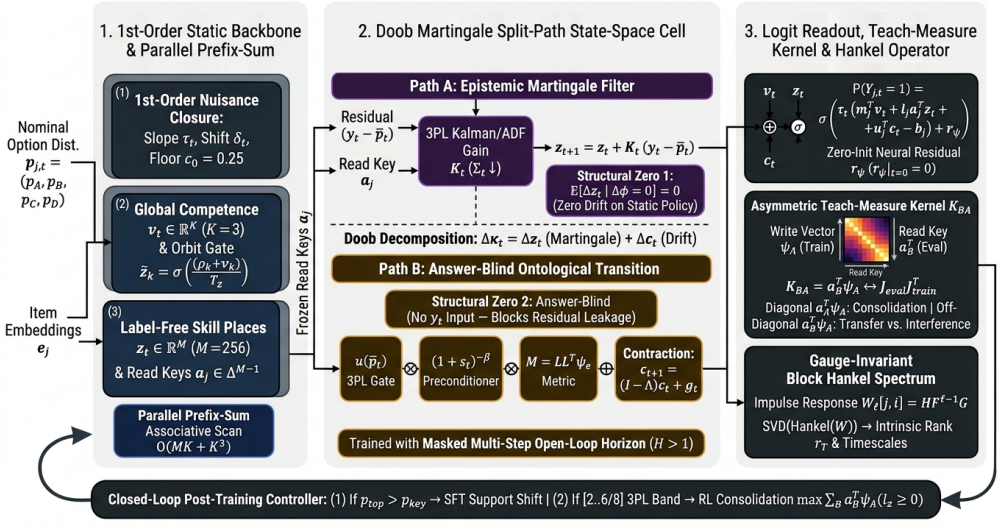
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3. Asymmetric Teach–Measure Kernel & Multi-Step Impulse Operator

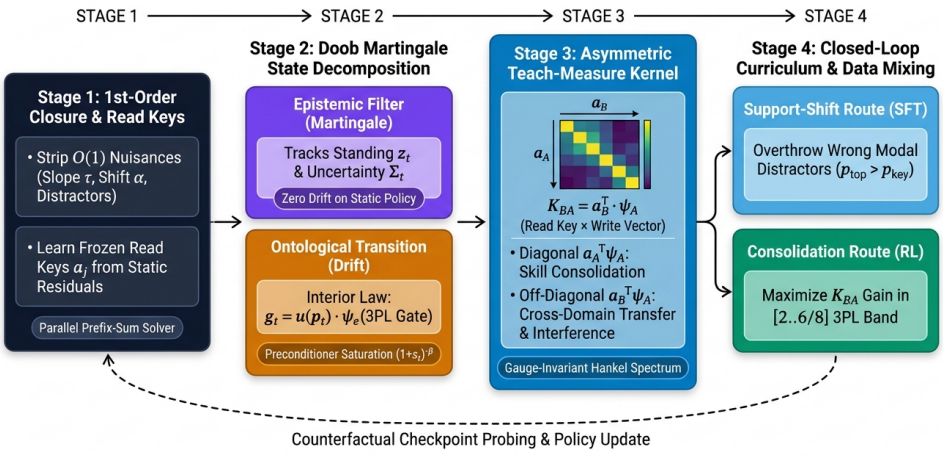
$$K_{BA} \;=\; \boldsymbol{a}_B^\top \boldsymbol{\psi}_A \;\longleftrightarrow\; \boldsymbol{J}_{\text{eval}}\,\boldsymbol{J}_{\text{train}}^\top, \qquad W_\ell[j,i] \;=\; \boldsymbol{H}\boldsymbol{F}^{\ell-1}\boldsymbol{G}$$

Unties **Read Key** $\boldsymbol{a}_B$ from **Write Vector** $\boldsymbol{\psi}_A$: diagonal K_{AA} measures skill consolidation, off-diagonal K_{BA} predicts cross-domain transfer vs. interference, and the Hankel spectrum of W_ℓ identifies the effective update rank.

State-Space Cell Architecture



Closed-Loop Curriculum and Data Mixing



Thank You!

First-Order Logit Expansion and Full Benchmark Results

First-Order Logit Expansion

For policy log-odds $z_j(\phi) = \operatorname{logit} p_j(\phi)$ under parameter update $\Delta\phi = \alpha_{\parallel} u_{\text{pop}} + u_{\perp}$:

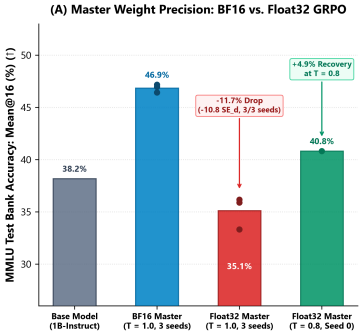
$$\Delta p_j \approx p_{0j}(1 - p_{0j})\left[\alpha_{\parallel} a_j^{\text{pop}} + J_j u_{\perp}\right]$$

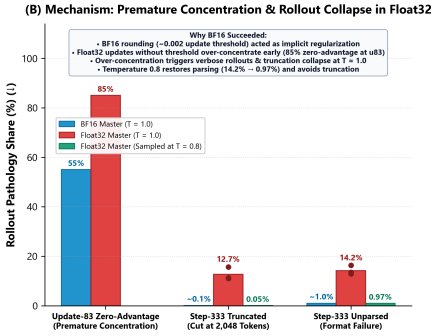
- $\operatorname{corr}(J_j u_{\perp}, a_j^{\text{pop}}) = \mathbf{0.072}$.
- Within-model variance $p_{0j}(1 - p_{0j})$ drives Δp_j ($R^2 = 0.354$), while a_j^{pop} has near-zero alignment ($r = +0.025$).

Disjoint MMLU Test Bank ($N = 1,472$)

Model / Arm	Draws	Pass@1	Pass@16	Maj@16	Order D
Base (Llama-3.2-1B)	—	38.2%	93.0%	47.6%	2.83%
CoT-SFT (70B Teacher)	0	43.9%	**96.6%**	**54.6%**	**1.88%**
GRPO Standard ($G = 5$)	5	46.9%	69.4%	47.5%	10.67%
Base Pass-Rate Baseline	1	47.2%	58.6%	47.4%	11.42%
2PL Unweighted ($\gamma = 0$)	1	46.9%	52.4%	47.1%	12.15%
2PL a_j-Weighted ($\gamma = 1$)	1	**47.5%**	58.7%	47.5%	11.80%
CoT-SFT @ T=0.2 (0 RL)	0	**47.5%**	87.9%	54.6%	1.95%
Two-Stage SFT → RL	1	46.7%	94.9%	54.4%	3.12%
Two-Stage [2..6/8 Band]	1	46.5%	94.2%	54.5%	2.98%

Numerical Precision Audit (BF16 vs. Float32)





(A) Float32 Master Weights Drop Accuracy by -11.7%

Removing BF16 rounding (~ 0.002 threshold) lowers $\mathrm{Pass@1}$ from $\mathbf{46.9}\% \rightarrow \mathbf{35.1}\%$ (partially recovered at $T = 0.8$).

(B) Premature Concentration in Float32

Exact Float32 updates over-concentrate early ($\mathbf{85}\%$ zero-variance rollouts vs. 55% in BF16), causing truncation (12.7%) and parse errors (14.2%).